```
burim@burim-virtual-machine:~$ sudo apt update
Ign:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Ign:2 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Ign:3 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Ign:4 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
Ign:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Ign:2 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Ign:3 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Ign:4 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
Ign:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Ign:2 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Ign:3 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Ign:4 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
Err:2 http://us.archive.ubuntu.com/ubuntu jammy InRelease
  Temporary failure resolving 'us.archive.ubuntu.com'
Err:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
  Temporary failure resolving 'security.ubuntu.com'
Err:3 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
  Temporary failure resolving 'us.archive.ubuntu.com'
Err:4 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
  Temporary failure resolving 'us.archive.ubuntu.com'
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
burim@burim-virtual-machine:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages have been kept back:
  cpp-11 gcc-11-base
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
burim@burim-virtual-machine:~$
```

```
burim@burim-virtual-machine:~
burim@burim-virtual-machine:~$ sudo reboot
```

```
burim@burim-virtual-machine:~$ sudo su root
[sudo] password for burim:
root@burim-virtual-machine:/home/burim#
```

The prompt changed from $ to #.

```
burim@burim-virtual-machine:~$ sudo su root
[sudo] password for burim:
Sorry, try again.
[sudo] password for burim:
root@burim-virtual-machine:/home/burim# useradd bobby
useradd: user 'bobby' already exists
root@burim-virtual-machine:/home/burim# adduser sally
adduser: The user `sally' already exists.
root@burim-virtual-machine:/home/burim# su - sally
sally@burim-virtual-machine:~$ useradd earl
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
sally@burim-virtual-machine:~$
```

The useradd command creates a user entry while adduser creates a home directory and asks for a password on creation. After swapping to sally the prompt changed from my name

to sally@burim. Trying to create a user as sally get denied as sally doesn't have sudo privilege. To have her create a new user I can give her sudo privilege.

```
root@burim-virtual-machine: /home/burim
[sudo] password for burim:
Sorry, try again.
[sudo] password for burim:
root@burim-virtual-machine:/home/burim# useradd bobby
useradd: user 'bobby' already exists
root@burim-virtual-machine:/home/burim# adduser sally
adduser: The user `sally' already exists.
root@burim-virtual-machine:/home/burim# su - sally
sally@burim-virtual-machine:~$ useradd earl
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
sally@burim-virtual-machine:~$ exit
logout
root@burim-virtual-machine:/home/burim# exit
exit
burim@burim-virtual-machine:~$ sudo userdel -r bobby
userdel: bobby mail spool (/var/mail/bobby) not found
userdel: bobby home directory (/home/bobby) not found
burim@burim-virtual-machine:~$ sudo userdel bobby
userdel: user 'bobby' does not exist
burim@burim-virtual-machine:~$ sudo su root
root@burim-virtual-machine:/home/burim# sudo userdel bobby
userdel: user 'bobby' does not exist
root@burim-virtual-machine:/home/burim#
```

```
burim@burim-virtual-machine:~$ sudo passwd sally
New password:
Retype new password:
passwd: password updated successfully
burim@burim-virtual-machine:~$
```

It is bad practice to stay logged in as root as a typo can ruin everything, my personal favorite is "sudo rm-rf / instead of sudo rm-rf./".

```
burim@burim-virtual-machine:~$ id
uid=1000(burim) gid=1000(burim) groups=1000(burim),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare)
burim@burim-virtual-machine:~$
```

Ubuntu belongs to sudo , adm, cdrom.

```
burim@burim-virtual-machine:~$ sudo usermod -aG sudo sally
burim@burim-virtual-machine:~$
```

```
burim@burim-virtual-machine:~$ sudo groupadd cybersec
burim@burim-virtual-machine:~$ sudo usermod -aG cybersec sally
burim@burim-virtual-machine:~$
```

```
burim@burim-virtual-machine:~$ id sally
uid=1002(sally) gid=1002(sally) groups=1002(sally),27(sudo),1003(cybersec)
burim@burim-virtual-machine:~$
```

The groups sally is in can be checked with groups or id.

```
burim@burim-virtual-machine:~$ mkdir lab1
burim@burim-virtual-machine:~$ ls -ld lab1
drwxrwxr-x 2 burim burim 4096 Mar  2 18:51 lab1
burim@burim-virtual-machine:~$
```

The permissions for the owner and group is read write and execute while others have execute.

```
burim@burim-virtual-machine:~$ nano helloWorld.sh
burim@burim-virtual-machine:~$ chmod +x helloWorld.sh
burim@burim-virtual-machine:~$
```

```
burim@burim-virtual-machine:~$ ls -l helloWorld.sh
-rwxrwxr-x 1 burim burim 19 Mar  2 18:53 helloWorld.sh
burim@burim-virtual-machine:~$
```

```
burim@burim-virtual-machine:~$ chmod g+wx helloWorld.sh
burim@burim-virtual-machine:~$ getfacl
Usage: getfacl [-aceEsRLPtpndvh] file ...
Try `getfacl --help' for more information.
burim@burim-virtual-machine:~$ getfacl helloWorld.sh
# file: helloWorld.sh
# owner: burim
# group: burim
user::rwx
group::rwx
other::r-x

burim@burim-virtual-machine:~$
```

```
burim@burim-virtual-machine:~$ setfacl -m u:sally:rw helloWorld.sh
burim@burim-virtual-machine:~$
```